

requests the moment there are two (*DOSPageCount*) requests in one (*DOSPageInterval*) second from the same host. If you ran the commands I have included earlier to test the configuration, you will have seen that we only get two or three cases of 'HTTP status code 200' before we start getting 'HTTP status code 403'. That happens because we hit the threshold in the first few requests.

Those numbers aren't very practical, and your specific requirements will be different, depending on how busy your website is. You will have to account for how long your pages take to load and how frequently you expect your users to try to reload any page. A *DOSPageCount* of 50 and *DOSPageInterval* of 20 seem very generous for the purposes of our article and for testing.

Let's change those numbers and run the *while* loop again to see how the behaviour changes. You will have to restart the Apache server with the following command:

```
sudo service apache2 restart
```

When you run the *while* loop, you may have to change the number of iterations to something close to 100 so that you can emulate an attack with the changed *mod_evasive* configuration:

```
count=1
while [ $count -le 100 ]
do
    curl -o /dev/null -s -w "%{http_
code}\n" http://localhost
    count=$(( $count + 1 ))
done
```

You may also want to adjust the *DOSBlockingInterval* for single site configuration, which defaults to 10 seconds before it unblocks the client in question again. If you see a large number of incoming brute force

attacks, you may want to increase it to something like 60 seconds to discourage the attackers.

The *DOSSiteCount* and *DOSSiteInterval* parameters work like *DOSPageCount* and *DOSPageInterval*, but site-wide instead of individual pages, to provide you further flexibility about when they start classifying the incoming traffic as an attack.

The *DOSSystemCommand* allows you to run commands to initiate further action in case the *mod_evasive* module detects an attack. You can use this parameter to invoke a script that can perform a firewall rule modification or raise a ticket in your ITSM tool to generate a security incident. This is apart from the notifications that you can send to the email address defined for the *DOSEmailNotify* parameter.

You can also use *DOSWhitelist* to whitelist certain IP addresses or ranges that will never be denied responses from your Web server. You can use this parameter multiple times to accommodate multiple IP addresses or ranges.

An example of when this parameter can be used would be a large customer for your SaaS application or a big client that you expect to send heavy traffic to your Web servers.

Augmenting *mod_evasive* for more security

Busier, larger and mission-critical Web servers will need additional security measures to fend off DDoS attacks, especially with the kind of scale the attackers can engineer with a botnet purchased from the deep corners of the Dark Web. *mod_evasive* can stand

on its own for simpler DDoS attacks but will fall short when attacks do not target the Web server directly or when launched with botnets, as such attacks would either not arrive at the Web server or may escape detection altogether by *mod_evasive* due to the extremely distributed nature of such attacks.

In such cases, augmenting web server security with host based firewalls, such as IPTables, and network firewalls becomes imperative. You can use the *DOSSystemCommand* directive to integrate *mod_evasive* as another signal for these firewalls to strengthen the detection and prevention capabilities. For example, you can use a script to run, which can perform any or all of the following:

- Create an incident for the security team to start investigating the traffic patterns.
- Alter the *localhost* firewall rules to tighten the host security, such as 'rate limit' connections from malicious IP addresses to limit the impact of a DDoS.
- Initiate processes to perform such actions on other services and hosts.

If you are running a Web server that attracts traffic, security needs to be your priority and the *mod_evasive* module for the Apache Web server is a good tool in your kit. This module works well for the purpose and scales with your Apache Web server; it is meant to be your last line of defence – when the unwanted traffic is hitting your Web host. Please be aware that a DDoS attack is just one of the many types of security threats that you will encounter once you are out there. So a comprehensive network perimeter firewall along with other security practices should be high on the list of things that you want to implement. **END** 

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